3/10/26 Sat

17/10/26.

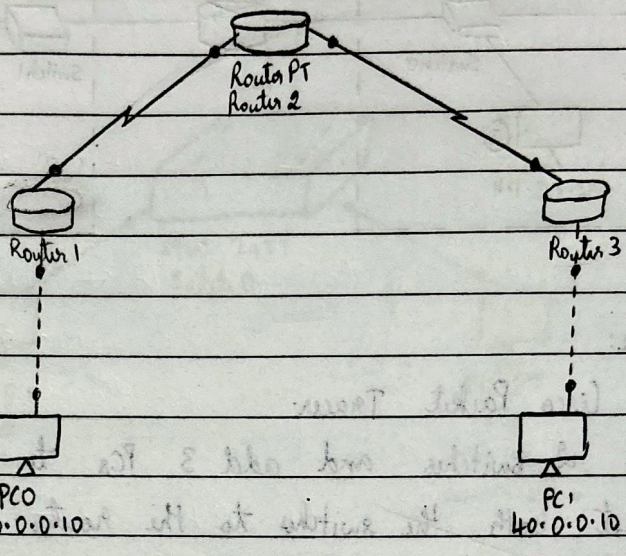
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1-Lab

Lab-2

Configure the IP Address to router in packet tracer. Explore the following messages: ping responses, destination unreachable, request timed out, reply



Procedure

1. Configuration - Open cisco packet tracer and add 3 routers and 2 end devices and test with PDU (pinging) Configure end devices with GUI, including subnet mask and default gateway, and CLI for 3 routers.

2. Configuration Commands

→ PC0		→ PC1	
IP Address	10.0.0.10	IP Address	40.0.0.10
Subnet Mask	255.0.0.0	Subnet Mask	255.0.0.0
Default Gateway	10.0.0.1	Default Gateway	40.0.0.1

R1

enable

Configure terminal

interface fastEthernet 0/0

ip address 10.0.0.1 255.0.0.0

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no shutdown
exit
interface serial 2/0
ip address 20.0.0.1 255.0.0.0
clock rate 64000
no shutdown
exit
ip route 40.0.0.0 255.0.0.0 20.0.0.2
end

→ R2
enable
configure terminal
interface serial 2/0
ip address 20.0.0.2 255.0.0.0
no shutdown
exit
interface serial 3/0
ip address 30.0.0.1 255.0.0.0
clock rate 64000
no shutdown
exit
ip route 10.0.0.0 255.0.0.0 20.0.0.1
ip route 40.0.0.0 255.0.0.0 30.0.0.2
end

→ R3
enable
configure terminal
interface serial 2/0
ip address 30.0.0.2 255.0.0.0
no shutdown
exit
ip route 10.0.0.0 255.0.0.0 30.0.0.1
end

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interface fastEthernet 0/0
ip address 40.0.0.1 255.0.0.0
no shutdown
exit
ip route 10.0.0.0 255.0.0.0 30.0.0.1
end

Test Connectivity:

Test1: PC0 → R1 (On PC0)
ping 10.0.0.1
Reply from 10.0.0.1 : bytes = 32 time=2ms TTL = 255
:

Test2: R1 → R2 (On R1)
ping 20.0.0.2
Sending 5, 100-byte ICMP Echos to 20.0.0.2, timeout is 2 se
!!!!!
Success Rate is 100 percent (5/5), round-trip min/avg/max = 1/3/5 ms

Test3: R2 → R3 (On R2)
ping 30.0.0.2
Sending 5, 100-byte ICMP Echos to 30.0.0.2, timeout is 2 seconds
!!!!!
Success Rate is 100 percent (5/5), round-trip min/avg/max = 2/2/3 ms

Test4: R3 → PC1 (On R3)
ping 40.0.0.10
Sending 5, 100-byte ICMP Echos to 40.0.0.10, timeout is 2 seconds
.!!!!
Success Rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

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Final PC0 → PC1 Test

On PC0:

ping 40.0.0.10

Pinging 40.0.0.10 with 32 bytes of data:

Reply from 40.0.0.10: bytes = 32 time = 5ms TTL = 125
Reply from 40.0.0.10 bytes = 32 time = 7ms TTL = 125
Reply from 40.0.0.10 bytes = 32 time = 9ms TTL = 125
Reply from 40.0.0.10 bytes = 32 time = 5ms TTL = 125

Pinging Statistics for 40.0.0.10:
Packets: Sent = 4, Recieved = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 5ms, Maximum = 9ms, Average = 6ms

Request Timed Out:

Temporarily disable R3's connection to PC1:
On R3:
configure terminal
interface fastEthernet 0/0
shutdown

Then, On PC1:
ping 40.0.0.10

Pinging 40.0.0.10 with 32 bytes of data:

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Reply from 30.0.0.2: Destination host unreachable
Reply from 30.0.0.2: Destination host unreachable
Reply from 30.0.0.2: Destination host unreachable
Reply from 30.0.0.2: Destination host unreachable

Ping Statistics for 40.0.0.10:
Packets: Sent = 4, Recieved = 0, Lost = 4 (100% Loss)

Later: (On R3)
R3 (config-if)# no shutdown

Destination Unreachable

Temporarily remove R1's route:
On R1:
configure terminal
no ip route 40.0.0.0 255.0.0.0 20.0.0.2

Then on PC1:
ping 40.0.0.10

Pinging 40.0.0.10 with 32 bytes of data:

Reply from 10.0.0.1: Destination host unreachable
Request timed out
Reply from 10.0.0.1: Destination host unreachable
Reply from 10.0.0.1: Destination host unreachable

Ping Statistics for 40.0.0.10:
Packets: Sent = 4, Recieved = 0, Lost = 4 (100% Loss)